Additional Exercises 2

Partial Derivatives & Differentiability

Mirrors: List 4 (exercises 1–5) | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Find the first-order partial derivatives

(a) ★ $z = x^3 + 3x^2y - y^3$

(b) ★ $z = \ln(x^2 + y^2)$

(c) ★ $z = \arctan\left(\frac{y}{x}\right)$

(d) ★ $z = \cos(ax - by)$

(e) ★★ $u = xe^{-yx}$

(f) ★★ $u = \frac{y}{x} + \frac{z}{y} + \frac{x}{z}$

(g) ★★ $z = \frac{xy}{x - y}$

(h) ★★ $u = \arcsin\left(\frac{y}{x}\right)$

(i) ★★★ $u = \sin^2(x + y) - \sin^2 x - \sin^2 y$

(j) ★★★ $z = \ln\left(\sqrt[3]{\frac{1}{x}} - \sqrt[3]{\frac{1}{y}}\right)$

Exercise 2: Calculate the partial derivatives at $(0, 0)$ from the definition

Do not use the standard differentiation rules — use the limit definition directly.

(a) ★ $f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$

(b) ★★ $f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$

(c) ★★ $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

(d) ★★ $f(x,y) = \begin{cases} x^2 \sin\dfrac{1}{y} + y^2 \sin\dfrac{1}{x}, & x \neq 0,\ y \neq 0, \\ 0, & \text{otherwise}. \end{cases}$

(e) ★★★ $f(x,y) = \begin{cases} \dfrac{x^3y - xy^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

Exercise 3: Verify the given identity

(a) ★★ $z = \ln(\sqrt{x} + \sqrt{y})$. Verify: $x\,\dfrac{\partial z}{\partial x} + y\,\dfrac{\partial z}{\partial y} = \dfrac{1}{2}$.

(b) ★★ $u = e^{x/t^2}$. Verify: $2x\,\dfrac{\partial u}{\partial x} + t\,\dfrac{\partial u}{\partial t} = 0$.

(c) ★★ $u = f(x^2+y^2+z^2)$, f differentiable. Verify:

$$y\,\frac{\partial u}{\partial x} - x\,\frac{\partial u}{\partial y} = 0.$$

(d) ★★★ $u = \sqrt{x^2+y^2+z^2}$. Verify: $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2 + \left(\dfrac{\partial u}{\partial z}\right)^2 = 1$.

(e) ★★★ $u = f\left(\dfrac{x}{y}, \dfrac{y}{z}\right)$, f differentiable. Show that $x\,\dfrac{\partial u}{\partial x} + y\,\dfrac{\partial u}{\partial y} + z\,\dfrac{\partial u}{\partial z} = 0$.

Exercise 4: Check differentiability at $(0,0)$

For each function: (1) compute $f_x(0,0)$, $f_y(0,0)$ from the limit definition; (2) check whether $\lim_{\Delta\rho\to 0}[\Delta f - f_x\Delta x - f_y\Delta y]/\Delta\rho = 0$.

(a) ★★ $f(x,y) = \begin{cases} \dfrac{x^3+y^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

(b) ★★★ $f(x,y) = \begin{cases} (x^2+y^2)\cos\dfrac{\pi}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

(c) ★★★ $f(x,y) = \sqrt[3]{xy}$ at $(0,0)$.

(a) Find $f_x(0,0)$ and $f_y(0,0)$.
(b) Are f_x, f_y continuous at $(0,0)$?
(c) Is f differentiable at $(0,0)$?
(Hint: (b) and (c) have different answers.)

(d) ★★★ $f(x,y) = \begin{cases} \dfrac{x^2y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

Are the partial derivatives continuous at $(0,0)$? Is f differentiable?

Exercise 5: Tangent plane and normal line

Recall: tangent plane to $z = f(x,y)$ at (a,b): $z - f(a,b) = f_x(a,b)(x-a) + f_y(a,b)(y-b)$.

(a) ★ $z = x^2 + y^2$ at the point $(1,1,2)$.

(b) ★★ $z = e^x \sin y$ at the point $(0, \pi/2, 1)$.

(c) ★★ $z = \ln(x^2 + y^2)$ at the point $(1,0,0)$.

(d) ★★ $z = \sqrt{x^2 + y^2}$ at the point $(3,4,5)$.

(e) ★★★ $x^2 + y^2 + z^2 = 14$ at the point $(1,2,3)$.
(Level surface form — use ∇F.)

(f) ★★★ Find the points on the surface $z = x^2 - xy + y^2 + 1$ where the tangent plane is parallel to the plane $2x + 4y - z = 5$.

★ *Standard differentiation rules* ★★ *Definition at a point or chain rule* ★★★ *Full differentiability criterion or level surface gradient.*